Asteria University of Engineering and Applied Sciences

MEng programs, advanced study, thesis, and graduation requirements

Purpose and award

This fictional catalog is a synthetic test document for CampusGPT. Asteria University of Engineering and Applied Sciences is invented; all rules and program information are sample data.

The Master of Engineering (MEng) is a two-year, full-time postgraduate degree organized into four semesters. The standard minimum is 80 credits. An approved part-time pathway may take up to four years.

Students choose either the thesis pathway or the applied engineering project pathway. Both pathways require advanced coursework, research or project ethics, a final examination, and a supervised engineering contribution.

Admission and preparation

Applicants normally hold a recognized BEng or equivalent four-year engineering degree in a relevant discipline, with a minimum GPA of 2.75 out of 4.00 or equivalent. Departments may require prerequisite study, a portfolio, an interview, or evidence of research preparation.

Applicants from a different engineering discipline may be admitted conditionally and assigned up to 12 credits of prerequisite coursework. Prerequisite credits do not count toward the 80-credit MEng graduation minimum.

The graduate application includes official transcripts, a statement of purpose, two academic or professional references, a current resume, and any department-specific materials.

Credit requirements and common core

The 80-credit MEng minimum consists of 32 credits of advanced required courses, 16 credits of technical electives, 8 credits of research methods and engineering ethics, and 24 credits of thesis or applied project work.

All students complete Graduate Research Methods, Engineering Ethics and Responsible Innovation, and a graduate seminar. Students must maintain a cumulative GPA of at least 3.00 out of 4.00. A grade below B in a required course may require a repeat or approved replacement.

The thesis pathway registers for Thesis Research I and II, submits a written thesis, and passes an oral defense. The applied project pathway registers for Engineering Project I and II, submits a validated design or implementation report, and passes a panel examination.

MEng specializations by department

Civil and Environmental Engineering: Structural Systems, Sustainable Infrastructure, Water Resources, and Environmental Process Design.

Computer and Software Engineering: Artificial Intelligence Systems, Secure Software, Distributed Computing, and Cyber-Physical Systems.

Electrical and Electronic Engineering: Power and Energy Systems, Embedded Systems, Signal Processing, and Communication Networks.

Mechanical and Aerospace Engineering: Robotics, Thermal Systems, Advanced Manufacturing, and Aerospace Systems.

Chemical and Materials Engineering: Process Systems, Advanced Materials, Sustainable Chemical Production, and Materials Characterization.

Industrial and Biomedical Engineering: Operations Research, Health Systems Engineering, Medical Devices, and Human-Centered Design.

Four-semester study plan

Semester 1: Graduate Research Methods, Advanced Mathematics for Engineers, Departmental Advanced Core I, and Graduate Seminar.

Semester 2: Engineering Ethics and Responsible Innovation, Departmental Advanced Core II, Technical Elective I, and Thesis Proposal or Project Definition.

Semester 3: Departmental Advanced Core III, Technical Elective II, Thesis Research I or Engineering Project I, and supervisor progress review.

Semester 4: Thesis Research II or Engineering Project II, final report or thesis, oral defense or panel examination, and final graduate seminar presentation.

Supervision, examination, and research conduct

Each student has a primary supervisor approved by the department. A thesis committee includes the supervisor and at least two additional qualified examiners, with one examiner external to the immediate project team.

A thesis proposal is normally approved by the end of Semester 2. The final thesis must document the problem, related work, methods, results, limitations, and references. The oral defense is assessed as pass, pass with revisions, or fail.

Applied projects must define engineering requirements, assess technical and societal risk, validate the proposed solution, document reproducible results, and present the work to a departmental panel.

All research must follow institutional ethics approval, data management, intellectual property, laboratory safety, and academic integrity policies.

MEng completion checklist

Earn at least 80 credits, including all common core and approved specialization requirements.

Maintain a cumulative GPA of at least 3.00 out of 4.00 and resolve incomplete grades.

Complete at least 24 credits of thesis or applied project work.

For the thesis pathway, submit the approved thesis and pass the oral defense; for the applied pathway, submit the final report and pass the panel examination.

Receive supervisor and department approval, submit the final corrected document to the Graduate Studies repository, and complete the graduation application.

Receive Registrar, library, laboratory, and financial clearance before the degree is conferred.